

2. Show that

$$\int_0^{\infty} \frac{8x - 12}{(2x^2 + 3)(x + 1)} dx = \ln k$$

where k is a rational number to be found.

(7)

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2. (a) Explain why $\int_1^{\infty} \frac{1}{x(2x+5)} dx$ is an improper integral. (1)

(b) Prove that

$$\int_1^{\infty} \frac{1}{x(2x+5)} dx = a \ln b$$

where a and b are rational numbers to be determined.

(6)

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- (2)

- (3)

- (1)

- (1)

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5. (i) Evaluate the improper integral

$$\int_1^{\infty} 2e^{-\frac{1}{2}x} dx \quad (3)$$

- (ii) The air temperature, $\theta^\circ\text{C}$, on a particular day in London is modelled by the equation

$$\theta = 8 - 5 \sin\left(\frac{\pi}{12}t\right) - \cos\left(\frac{\pi}{6}t\right) \quad 0 \leq t \leq 24$$

where t is the number of hours after midnight.

- (a) Use calculus to show that the mean air temperature on this day is 8°C , according to the model. (3)

Given that the actual mean air temperature recorded on this day was higher than 8°C ,

- (b) explain how the model could be refined. (1)

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9. (a) Use a hyperbolic substitution and calculus to show that

$$\int \frac{x^2}{\sqrt{x^2 - 1}} dx = \frac{1}{2} \left[x\sqrt{x^2 - 1} + \operatorname{arcosh} x \right] + k$$

where k is an arbitrary constant.

(6)

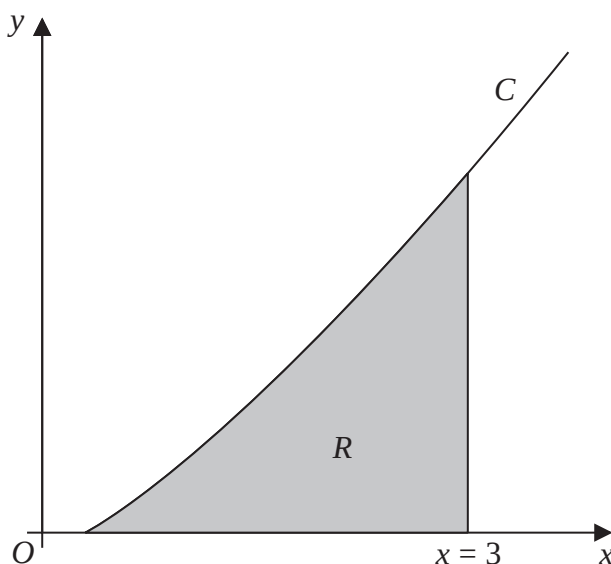


Figure 1

Figure 1 shows a sketch of part of the curve C with equation

$$y = \frac{4}{15} x \operatorname{arcosh} x \quad x \geq 1$$

The finite region R , shown shaded in Figure 1, is bounded by the curve C , the x -axis and the line with equation $x = 3$

- (b) Using algebraic integration and the result from part (a), show that the area of R is given by

$$\frac{1}{15} \left[17 \ln(3 + 2\sqrt{2}) - 6\sqrt{2} \right]$$

(5)



6. (a) Express as partial fractions

$$\frac{2x^2 + 3x + 6}{(x+1)(x^2 + 4)}$$

(3)

(b) Hence, show that

$$\int_0^2 \frac{2x^2 + 3x + 6}{(x+1)(x^2 + 4)} dx = \ln(a\sqrt{2}) + b\pi$$

where a and b are constants to be determined.

(4)

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8. (a) Given

$$z^n + \frac{1}{z^n} = 2 \cos n\theta \quad n \in \mathbb{N}$$

show that

$$32 \cos^6 \theta \equiv \cos 6\theta + 6 \cos 4\theta + 15 \cos 2\theta + 10 \quad (5)$$

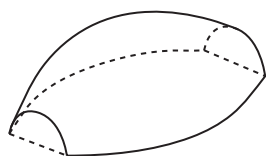


Figure 1

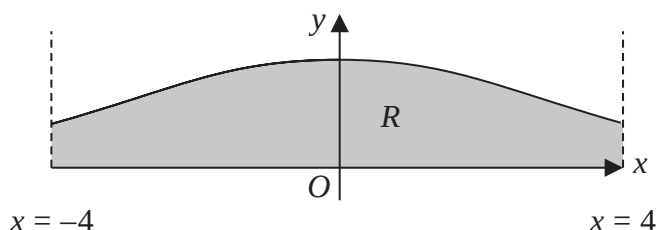


Figure 2

Figure 1 shows a solid paperweight with a flat base.

Figure 2 shows the curve with equation

$$y = H \cos^3 \left(\frac{x}{4} \right) \quad -4 \leq x \leq 4$$

where H is a positive constant and x is in radians.

The region R , shown shaded in Figure 2, is bounded by the curve, the line with equation $x = -4$, the line with equation $x = 4$ and the x -axis.

The paperweight is modelled by the solid of revolution formed when R is rotated 180° about the x -axis.

Given that the maximum height of the paperweight is 2 cm,

(b) write down the value of H . (1)

(c) Using algebraic integration and the result in part (a), determine, in cm^3 , the volume of the paperweight, according to the model. Give your answer to 2 decimal places.

[Solutions based entirely on calculator technology are not acceptable.] (5)

(d) State a limitation of the model. (1)



9. (i) (a) Explain why $\int_0^{\infty} \cosh x \, dx$ is an improper integral.

(1)

(b) Show that $\int_0^{\infty} \cosh x \, dx$ is divergent.

(3)

(ii) $4 \sinh x = p \cosh x$ where p is a real constant

Given that this equation has real solutions, determine the range of possible values for p

(2)

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2. (a) Write $x^2 + 4x - 5$ in the form $(x + p)^2 + q$ where p and q are integers.

(1)

(b) Hence use a standard integral from the formula book to find

$$\int \frac{1}{\sqrt{x^2 + 4x - 5}} dx$$

(2)

(c) Determine the mean value of the function

$$f(x) = \frac{1}{\sqrt{x^2 + 4x - 5}} \quad 3 \leq x \leq 13$$

giving your answer in the form $A \ln B$ where A and B are constants in simplest form.

(3)

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2.

$$f(x) = \tanh^{-1}\left(\frac{3-x}{6+x}\right) \quad |x| < \frac{3}{2}$$

(a) Show that

$$f'(x) = -\frac{1}{2x+3} \quad (4)$$

(b) Hence determine $f''(x)$ (1)(c) Hence show that the Maclaurin series for $f(x)$, up to and including the term in x^2 , is

$$\ln p + qx + rx^2$$

where p , q and r are constants to be determined. (3)



3. (a) Explain why

$$\int_{\frac{4}{3}}^{\infty} \frac{1}{9x^2 + 16} dx$$

is an improper integral.

(1)

(b) Show that

$$\int_{\frac{4}{3}}^{\infty} \frac{1}{9x^2 + 16} dx = k\pi$$

where k is a constant to be determined.

(4)



9.

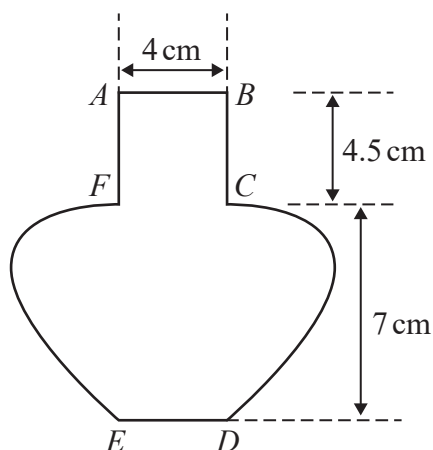


Figure 1

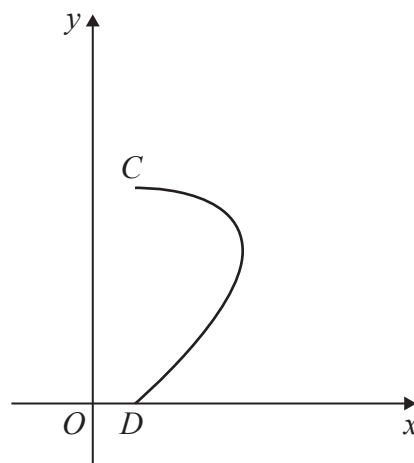


Figure 2

Figure 1 shows the central vertical cross-section $ABCDEFA$ of a vase together with measurements that have been taken from the vase.

The horizontal cross-section between AB and FC is a circle with diameter 4 cm.

The base of the vase ED is horizontal and the point E is vertically below F and the point D is vertically below C .

Using these measurements, the curve CD is modelled by the parametric equations

$$x = a + 3 \sin 2t \quad y = b \cos t \quad 0 \leq t \leq \frac{\pi}{2}$$

where a and b are constants and O is the fixed origin, as shown in Figure 2.

- (a) Determine the value of a and the value of b according to the model. (2)
- (b) Using algebraic integration and showing all your working, determine, according to the model, the volume of the vase, giving your answer to the nearest cm^3 (7)
- (c) State a limitation of the model. (1)



3.

$$f(x) = \frac{1}{\sqrt{4x^2 + 9}}$$

- (a) Using a substitution, that should be stated clearly, show that

$$\int f(x)dx = A \sinh^{-1}(Bx) + c$$

where c is an arbitrary constant and A and B are constants to be found.

(4)

- (b) Hence find, in exact form in terms of natural logarithms, the mean value of $f(x)$ over the interval $[0, 3]$.

(2)

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8.

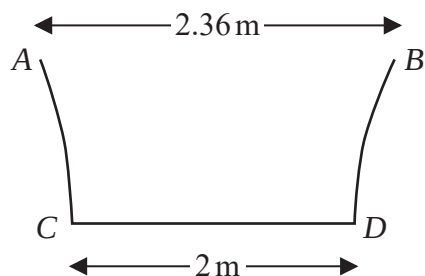


Figure 1

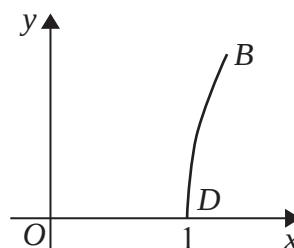


Figure 2

Figure 1 shows the central vertical cross section $ABCD$ of a paddling pool that has a circular horizontal cross section. Measurements of the diameters of the top and bottom of the paddling pool have been taken in order to estimate the volume of water that the paddling pool can contain.

Using these measurements, the curve BD is modelled by the equation

$$y = \ln(3.6x - k) \quad 1 \leq x \leq 1.18$$

as shown in Figure 2.

(a) Find the value of k .

(1)

(b) Find the depth of the paddling pool according to this model.

(2)

The pool is being filled with water from a tap.

(c) Find, in terms of h , the volume of water in the pool when the pool is filled to a depth of h m.

(5)

Given that the pool is being filled at a constant rate of 15 litres every minute,

(d) find, in cm h^{-1} , the rate at which the water level is rising in the pool when the depth of the water is 0.2 m.

(3)

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5. (a)

$$y = \tan^{-1} x$$

Assuming the derivative of $\tan x$, prove that

$$\frac{dy}{dx} = \frac{1}{1+x^2} \quad (3)$$

$$f(x) = x \tan^{-1} 4x$$

(b) Show that

$$\int f(x) dx = Ax^2 \tan^{-1} 4x + Bx + C \tan^{-1} 4x + k$$

where k is an arbitrary constant and A , B and C are constants to be determined.

(5)

(c) Hence find, in exact form, the mean value of $f(x)$ over the interval $\left[0, \frac{\sqrt{3}}{4}\right]$ (2)

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7.

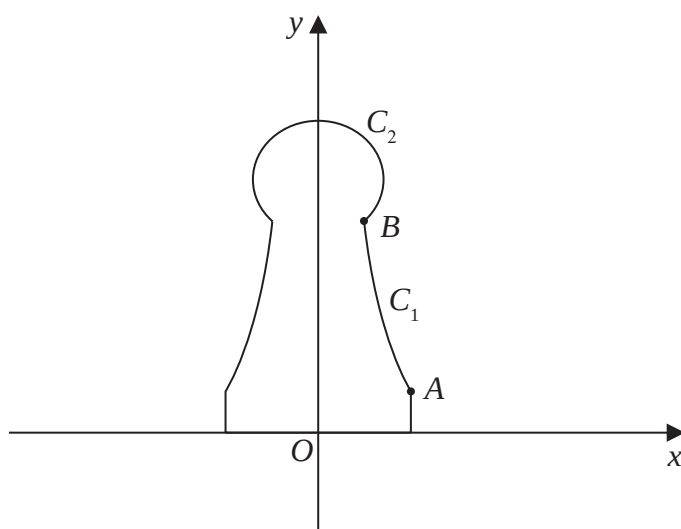


Figure 1

A student wants to make plastic chess pieces using a 3D printer. Figure 1 shows the central vertical cross-section of the student's design for one chess piece. The plastic chess piece is formed by rotating the region bounded by the y -axis, the x -axis, the line with equation $x = 1$, the curve C_1 and the curve C_2 through 360° about the y -axis.

The point A has coordinates $(1, 0.5)$ and the point B has coordinates $(0.5, 2.5)$ where the units are centimetres.

The curve C_1 is modelled by the equation

$$x = \frac{a}{y + b} \quad 0.5 \leq y \leq 2.5$$

(a) Determine the value of a and the value of b according to the model.

(2)

The curve C_2 is modelled to be an arc of the circle with centre $(0, 3)$.

(b) Use calculus to determine the volume of plastic required to make the chess piece according to the model.

(9)

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3.

$$f(x) = \arcsin x \quad -1 \leq x \leq 1$$

- (a) Determine the first two non-zero terms, in ascending powers of x , of the Maclaurin series for $f(x)$, giving each coefficient in its simplest form.

(4)

- (b) Substitute $x = \frac{1}{2}$ into the answer to part (a) and hence find an approximate value for π

Give your answer in the form $\frac{p}{q}$ where p and q are integers to be determined.

(2)

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5. The curve C has equation

$$y = \arccos\left(\frac{1}{2}x\right) \quad -2 \leq x \leq 2$$

(a) Show that C has no stationary points.

(3)

The normal to C , at the point where $x = 1$, crosses the x -axis at the point A and crosses the y -axis at the point B .

Given that O is the origin,

(b) show that the area of the triangle OAB is $\frac{1}{54}(p\sqrt{3} + q\pi + r\sqrt{3}\pi^2)$ where p , q and r are integers to be determined.

(5)

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7. Solutions based entirely on graphical or numerical methods are not acceptable.

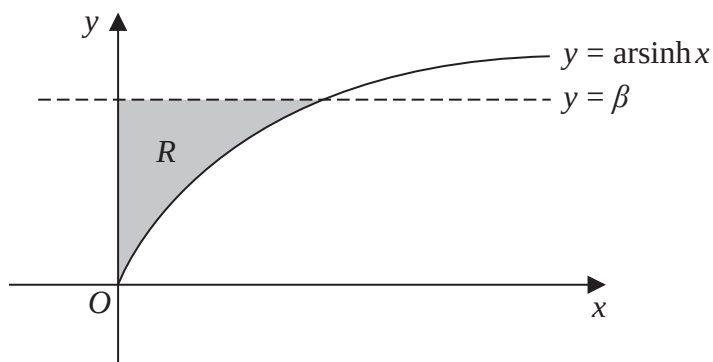


Figure 1

Figure 1 shows a sketch of part of the curve with equation

$$y = \operatorname{arsinh} x \quad x \geqslant 0$$

and the straight line with equation $y = \beta$

The line and the curve intersect at the point with coordinates (α, β)

Given that $\beta = \frac{1}{2} \ln 3$

(a) show that $\alpha = \frac{1}{\sqrt{3}}$ (3)

The finite region R , shown shaded in Figure 1, is bounded by the curve with equation $y = \operatorname{arsinh} x$, the y -axis and the line with equation $y = \beta$

The region R is rotated through 2π radians about the y -axis.

(b) Use calculus to find the exact value of the volume of the solid generated. (6)



5. (a) Given that

$$y = \arcsin x \quad -1 \leq x \leq 1$$

show that

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} \quad (3)$$

(b) $f(x) = \arcsin(e^x)$ $x \leq 0$

Prove that $f(x)$ has no stationary points.

(3)

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$$r = 1 + \tan \theta \qquad 0 \leq \theta < \frac{\pi}{3}$$

(a) Use differentiation to prove that the polar coordinates of A are $\left(2, \frac{\pi}{4}\right)$

(b) Use calculus to show that the exact area of R is $\frac{1}{2}(1 - \ln 2)$

(6)

$$y = \cosh^n x \quad n \geq 5$$

(a) (i) Show that

$$\frac{d^2 y}{dx^2} = n^2 \cosh^n x - n(n-1) \cosh^{n-2} x \quad (4)$$

(ii) Determine an expression for $\frac{d^4 y}{dx^4}$

(2)

(b) Hence determine the first three non-zero terms of the Maclaurin series for y , giving each coefficient in simplest form.

(2)



1.

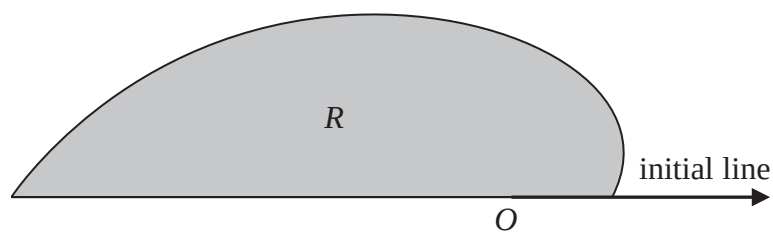
**Figure 1**

Figure 1 shows a sketch of the curve with polar equation

$$r = 2\sqrt{\sinh \theta + \cosh \theta} \quad 0 \leq \theta \leq \pi$$

The region R , shown shaded in Figure 1, is bounded by the initial line, the curve and the line with equation $\theta = \pi$

Use algebraic integration to determine the exact area of R giving your answer in the form $pe^q - r$ where p , q and r are real numbers to be found.

(4)

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2. (a) Write down the Maclaurin series of e^x , in ascending power of x , up to and including the term in x^3

(1)

- (b) Hence, without differentiating, determine the Maclaurin series of

$$e^{(e^x - 1)}$$

in ascending powers of x , up to and including the term in x^3 , giving each coefficient in simplest form.

(5)

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7.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

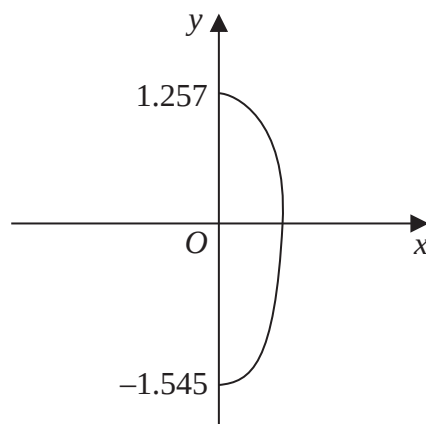


Figure 2

John picked 100 berries from a plant.

The largest berry picked was approximately 2.8 cm long.

The shape of this berry is modelled by rotating the curve with equation

$$16x^2 + 3y^2 - y \cos\left(\frac{5}{2}y\right) = 6 \quad x \geq 0$$

shown in Figure 2, about the y-axis through 2π radians, where the units are cm.

Given that the y intercepts of the curve are -1.545 and 1.257 to four significant figures,

- (a) use algebraic integration to determine, according to the model, the volume of this berry.

(6)

Given that the 100 berries John picked were then squeezed for juice,

- (b) use your answer to part (a) to decide whether, in reality, there is likely to be enough juice to fill a 200 cm^3 cup, giving a reason for your answer.

(2)

